

Risk Classification and Performance Reporting Model

Project Overview

The objective of this project was to simulate an operational reporting environment that enables leadership to identify performance risks before organizational targets are missed. Using SQL Server, I developed a risk classification model that tracks employee performance against monthly goals and provides actionable insights for decision-makers through dashboard reporting.

Data Preparation and Data Quality

The source dataset contained 1,000 employee records, including employee identifiers, job titles, departments, business units, hire dates, and exit dates.

During data validation, I discovered that the EEID field was reused for certain employees after termination, resulting in duplicate identifiers. To ensure reliable reporting and maintain referential integrity, I created a unique EmployeeID primary key and used it as the foundation for all table relationships.

This step improved data quality and ensured that reporting calculations were based on unique employee records.

Database Design

To support KPI and performance reporting, I designed a MonthlyPerformance table containing:

PerformanceID (Primary Key)

EmployeeID (Foreign Key)

ReportMonth

Target_Completions

Actual_Completions

SLA_Target

SLA_Achieved

Because the source data did not contain operational performance metrics, I generated twelve months of performance data for each employee, resulting in 12,000 performance records.

The SLA (Service Level Agreement) metrics were included to simulate contractual or operational performance standards commonly monitored within healthcare and business environments.

Risk Classification Methodology

Performance was measured using the following calculation:

Performance Score = Actual_Completions ÷ Target_Completions

Employees were classified into three categories:

Performance Score	Risk Status
95% or greater	On Track
90% to 94.99%	Medium Risk
Below 90%	High Risk

This methodology created a leading indicator that could be used to identify underperforming employees, programs, or departments before month-end performance targets were missed.

Analysis and Reporting

Using SQL queries and Power BI visualizations, performance data was aggregated by department and risk category to identify areas requiring management attention.

Figure 1 presents a department-level risk summary for January 2025. This analysis enables leadership to identify departments with elevated concentrations of high-risk employees and prioritize operational interventions where they are most needed.

Figure 1. Department-Level Performance Risk Summary for January 2025

Results Messages				
	Department	TotalEmployees	HighRiskEmployees	HighRiskPct
1	IT	241	106	44.000000000000
2	Accounting	96	42	43.800000000000
3	Human Resources	125	52	41.600000000000
4	Engineering	158	64	40.500000000000
5	Marketing	120	48	40.000000000000
6	Sales	140	55	39.300000000000
7	Finance	120	43	35.800000000000

Business Value

This reporting solution provides management with the ability to:

- Monitor organizational performance against monthly targets.
- Identify departments at elevated risk of missing performance goals.
- Track performance trends over time.
- Support proactive intervention before performance issues become larger operational concerns.
- Improve confidence in reporting through data-quality validation and structured database relationships.

Technical Skills Demonstrated

- SQL Server Database Design
- Data Cleaning and Validation
- Primary and Foreign Key Relationships
- KPI Development and Performance Measurement
- Risk Classification Modeling
- Data Aggregation and Reporting
- Business Intelligence and Dashboard Development
- Operational Performance Analysis
- Project Outcome

My project demonstrates the complete analytics lifecycle, from data preparation and quality validation through database design, KPI development, risk modeling, and executive reporting. The resulting solution transforms raw employee data into actionable business insights that support performance management and strategic decision-making.